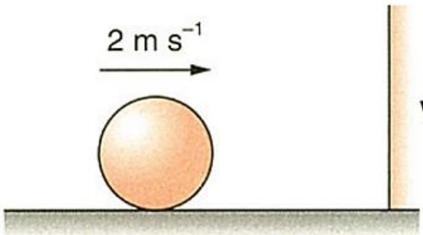
1. A ball moving at $2\ \mathrm{m\ s^{-1}}$ towards the right collides with a wall and rebounds at $1.5\ \mathrm{m\ s^{-1}}$ in the opposite direction. The mass of the ball is 0.8 kg. What is the change in momentum of the ball?

A. $0.4\ \mathrm{m\ s^{-1}}$ (towards right)
B. $0.4\ \mathrm{m\ s^{-1}}$ (towards left)
C. $2.8\ \mathrm{m\ s^{-1}}$ (towards right)
D. $2.8\ \mathrm{m\ s^{-1}}$ (towards left)



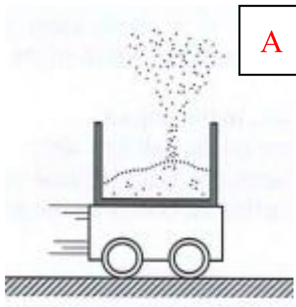
D

2. A trolley with a container at the top moves with uniform velocity to the right as shown. Sand is then continuously dropped into the container. What are the effects of the sand on the speed, the momentum and the kinetic energy of the trolley ?

	speed	momentum	kinetic energy
A.	decreased	unchanged	decreased
B.	decreased	unchanged	unchanged
C.	unchanged	unchanged	unchanged
D.	unchanged	increased	increased

By conservation of momentum in the horizontal direction (as there is no external horizontal force),
$m_{car}u_{car} = (m_{sand} + m_{car})v_{car}$
As sand is dropped into the container, m_{sand} increases, and so v_{car} decreases. Momentum of the car (including sand) remains constant.

K.E. of the car $= \frac{1}{2}(m_{sand} + m_{car})v_{car}^2$
$= \left(\frac{1}{2}m_{car}u_{car}\right)v_{car}$
$< \left(\frac{1}{2}m_{car}u_{car}\right)u_{car} = \frac{1}{2}m_{car}u_{car}^2$
$\therefore$ K.E. of the car decreases.



A

3. Two particles A and B of masses 2 kg and 1 kg respectively move in opposite directions. The initial velocity of A is 4 m s⁻¹ towards the right, while that of B is 2 m s⁻¹ towards the left. They collide head on. After the collision, the velocity of A becomes 1 m s⁻¹ towards the right. What would be the velocity of particle B ?

A. 2 m s⁻¹ towards the right
B. 3 m s⁻¹ towards the right
C. 4 m s⁻¹ towards the right
D. 6 m s⁻¹ towards the right

C

Take rightward direction as positive,

$m_A u_A + m_B u_B = m_A v_A + m_B v_B$

$(2)(4) + (1)(-2) = (2)(1) + (1)v_B \qquad \therefore\ v_B = 4\ \mathrm{m\ s^{-1}}$

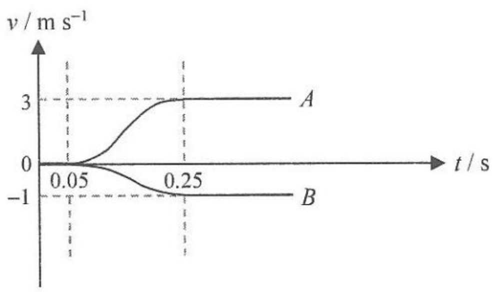
4. A 6000 kg van travelling at $90\ \mathrm{kmh^{-1}}$ collides head on with a 5000 kg stationary van. If the two vans stick together and move at the same velocity after collision, what is their velocity?

A. $11.1\ \mathrm{ms^{-1}}$
B. $13.6\ \mathrm{ms^{-1}}$
C. $30\ \mathrm{ms^{-1}}$
D. $49.1\ \mathrm{ms^{-1}}$

B

5. In an explosion, an object is blown into 2 pieces, A and B, which fly off in opposite directions. The mass of A is 0.3 kg. The graph shows the variation of velocity of A and B with time before and after the explosion. What is the mass of B and the estimated magnitude of the average net force acting on B during the explosion ?

	Mass of B	Magnitude of the average net force acting on B
A.	0.1 kg	0.4 N
B.	0.1 kg	0.5 N
C.	0.9 kg	3.6 N
D.	0.9 kg	4.5 N



D

① By Conservation of momentum : $m_A u_A + m_B u_B = m_A v_A + m_B v_B$

$(0) + (0) = (0.3)(3) + m_B(-1) \qquad \therefore\ m_B = 0.9$ kg

② By $F_B = \dfrac{mv - mu}{t} = \dfrac{(0.9)(-1) - 0}{(0.25 - 0.05)} = -4.5$ N

$\therefore$ Magnitude of average net force on B = 4.5 N

6. Two trolleys A and B, of masses 2 kg and 3 kg respectively, have the same momentum. What is the ratio of the kinetic energy of trolley A to that of trolley B ?

A. 2 : 3
B. 3 : 2
C. 4 : 9
D. 9 : 4

$m_A v_A = m_B v_B$

$\dfrac{m_A}{m_B} = \dfrac{v_B}{v_A} = \dfrac{2}{3}$

$\therefore \dfrac{\text{K.E.}_A}{\text{K.E.}_B} = \dfrac{\frac{1}{2} m_A {v_A}^2}{\frac{1}{2} m_B {v_B}^2} = \dfrac{m_A}{m_B}\left(\dfrac{v_A}{v_B}\right)^2 = \dfrac{2}{3}\left(\dfrac{3}{2}\right)^2 = \dfrac{3}{2}$

B

7. A rocket of mass 5000 kg is at rest in space. It then explodes and breaks into two parts P_1 and P_2 of mass 1000 kg and 4000 kg respectively. Find the ratio of the kinetic energy of P_1 to P_2.

A. 1 : 16
B. 1 : 64
C. 4 : 1
D. 16 : 1

As $KE = \frac{1}{2} m v^2 = \dfrac{(mv)^2}{m} \propto \dfrac{1}{m}$

KE of P_1 : KE of P_2 = mass of P_2 : mass of P_1 = 4000 : 1000 = 4 : 1

C

8. A basketball falls freely from rest and hits the ground. It then rebounds to 1/4 of its original height. Neglecting air resistance, which of the following statements about the basketball is/are correct ?

(1) Its kinetic energy just before collision is four times its kinetic energy just after collision.

(2) Its potential energy just before collision is four times its potential energy just after collision.

(3) The speed just before collision is two times the speed just after collision.

D

A. (1) only
B. (2) only
C. (2) & (3) only
D. (1) & (3) only

✓ (1) $PE = mgh$ $\therefore PE \propto h$

Rises by $\frac{1}{4}$ of its original height $\Rightarrow$ it has $\frac{1}{4}$ of its original PE at topmost point.

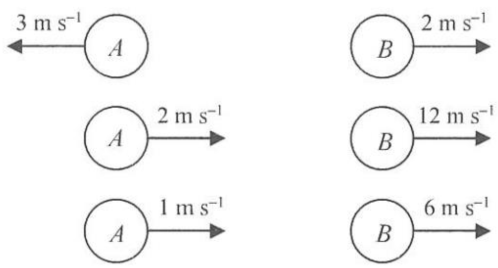
As loss of KE = gain in PE $\therefore$ KE at the lowest point has $\frac{1}{4}$ of its original value after collision.

× (2) Same level at these 2 instants $\Rightarrow$ same PE just before and just after collision

✓ (3) By $KE = \frac{1}{2} mv^2 \propto v^2$, the speed just before collision is double that just after collision.

9. Ball A and ball B of masses 2 kg and 1 kg respectively collide head-on as shown above. Which of the following diagram show(s) the possible result(s) after the collision ?

(1)
(2)
(3)



B

A. (1) only
B. (3) only
C. (1) & (2) only
D. (2) & (3) only

Check that whether the total momentum before and after collision is conserved or not.

Take the rightward direction as positive.

× (1) (2) (6) + (1) (−4) ≠ (2) (−3) + (1) (2)

× (2) (2) (6) + (1) (−4) ≠ (2) (2) + (1) (12)

✓ (3) (2) (6) + (1) (−4) = (2) (1) + (1) (6)

10. A cube is at rest on a rough inclined plane as shown. A bullet of mass $0.1\ kg$ is fired along the plane into the cube and is embedded in it. The mass of the cube is $2\ kg$. The plane is inclined at $30°$ to the horizontal. The velocity of the bullet is $80\ \text{ms}^{-1}$. After being hit by the bullet, the cube moves a distance of $0.7\ m$ up along the plane and then stops. What is the friction acting on the cube when it moves along the plane?

B

A. $7.21\ N$
B. $11.5\ N$
C. $14.5\ N$
D. $21.8\ N$

B By conservation of momentum,

$m_B u_B + 0 = (m_B + m_C)v$

$0.1 \times 80 = (0.1 + 2)v$

$v = 3.81\ \text{m s}^{-1}$

$\therefore$ The velocity of the cube is $3.81\ \text{m s}^{-1}$ just after the bullet is embedded in it.

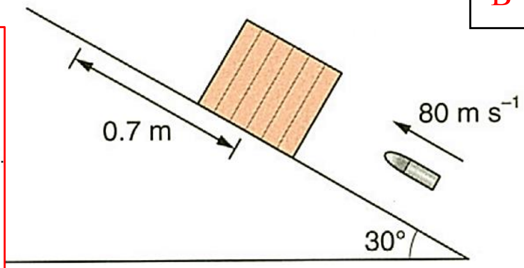
By conservation of energy,

loss in KE = gain in PE + work done against friction

$\frac{1}{2}(0.1 + 2)3.81^2 = (0.1 + 2)9.81(0.7 \sin 30°) + f(0.7)$

$f = 11.5\ \text{N}$

$\therefore$ The friction acting on the cube is 11.5 N.



11. George stands on trolley X which is at rest on a smooth horizontal surface. The masses of George and trolley X are respectively $70\ kg$ and $30\ kg$. Trolley Y of mass $50\ kg$ moves towards trolley X at $7\ \text{ms}^{-1}$ from the left as shown. When the trolleys are about to collide, George jumps horizontally with a velocity v towards trolley Y in order to avoid collision. What is the minimum value of v?

A

A. $1\ \text{ms}^{-1}$
B. $2.33\ \text{ms}^{-1}$
C. $3.5\ \text{ms}^{-1}$
D. $5\ \text{ms}^{-1}$

A With a minimum value of v, the two trolleys can just avoid collision and they move at the same final velocity v_F.

Take the moving direction of trolley Y as positive.

Consider trolley X. By conservation of momentum,

$0 = 70v + 30v_F$ ....(1)

Consider trolley Y. By conservation of momentum,

$50 \times 7 + 70v = (50 + 70)v_F$

$350 = 120v_F - 70v$ ....(2)

(1) + (2),

$350 = 150v_F$

$v_F = \frac{7}{3}\ \text{m s}^{-1}$

From (1), $v = \frac{-30v_F}{70} = \frac{-30 \times \frac{7}{3}}{70} = -1\ \text{m s}^{-1}$

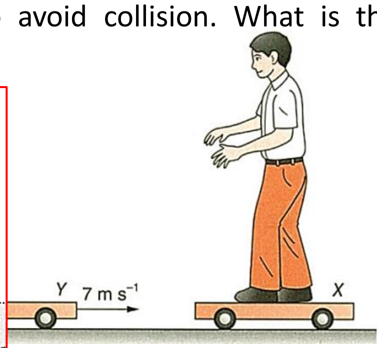
STRATEGY

Alternatively, we can consider the trolleys and George together to find v_F.

By conservation of momentum,

$0 + 50 \times 7 = (70 + 30 + 50)v_F$

$v_F = \frac{350}{150} = \frac{7}{3}\ \text{m s}^{-1}$

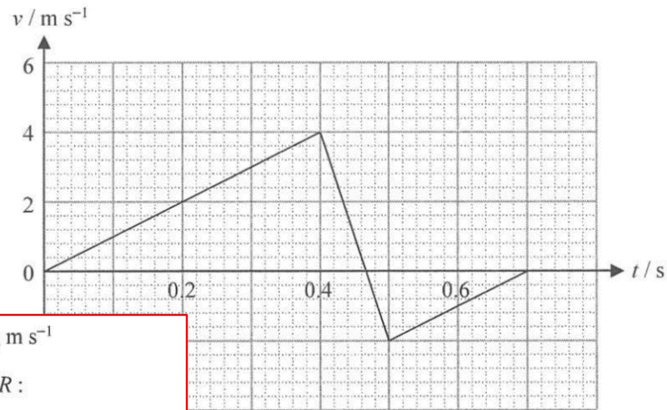


12. A ball of mass 0.2 kg is released from rest. It hits the ground and rebounds. The velocity-time graph of the ball is shown above. Which of the following statements are correct ?

(1) The magnitude of the change in momentum of the ball during the collision is 1.2 kg m s^{-1}.
(2) The magnitude of the average force acting on the ball by the ground during the collision is 12 N.
(3) There is mechanical energy loss during the collision.

B

A. (1) & (2) only
B. (1) & (3) only
C. (2) & (3) only
D. (1), (2) & (3)



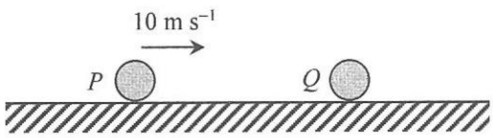
✓ (1) Change in momentum of the ball $= (0.2)(-2) - (0.2)(4) = -1.2\ \text{kg m s}^{-1}$

× (2) Average force acting on the ball by the ground is the normal reaction R :

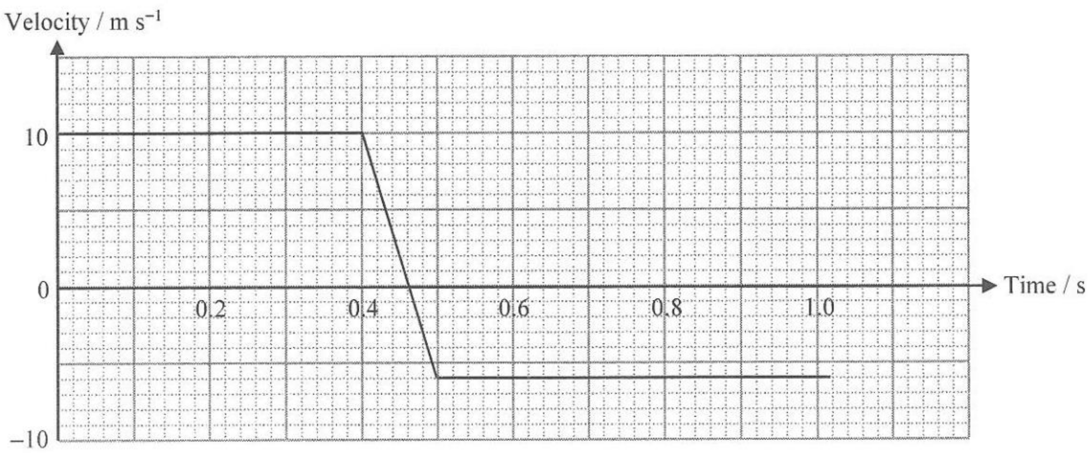
$R - mg = \frac{mv - mu}{t}$ $\quad \therefore\ R - (0.2)(9.81) = \frac{1.2}{0.1}$ $\quad \therefore\ R = 12 + 1.96 = 14.0\ \text{N}$

✓ (3) Since the speed after the collision is smaller, there is loss of KE during the collision.

13. (a)



A metal ball P of mass 0.5 kg moves with speed 10 m s^{-1} on smooth horizontal ground. It collides with a heavier metal ball Q, which is initially at rest. After collision, P moves backwards in the opposite direction. The figure below shows the variation of the velocity of P with time.



(i) Find

(1) the momentum of P before the collision,

(2) the change in momentum of P in the collision,

(3) the time of contact of P and Q,

(4) the average force acting on P during the collision. (6 marks)

(ii) Is the average force acting on Q during the collision equal in magnitude to that acting on P ? Explain briefly. (2 marks)

(iii) Comment on the following statement :

"Momentum and kinetic energy must be conserved in this collision." (4 marks)

(b) For safety reasons, the front and rear parts of cars should not be made of very strong material. Explain briefly. (2 marks)

(a) (i) (1) Momentum of P before collision $= 0.5 \times 10 = 5$ kg m s^{-1} [1]

(2) Change in momentum of P $= m\,v - m\,u$ [1]

$= (0.5) \times (-6) - (0.5) \times (10)$

$= -8$ kg m s^{-1} [1]

(3) Time of contact $= 0.1$ s [1]

(4) Average force on P $= \dfrac{m\,v - m\,u}{t} = \dfrac{-8}{0.1}$ [1]

$= -80$ N [1]

(ii) Yes! The average force acts on Q is equal in magnitude to that acting on P [1]

because the two forces are action and reaction pair according to Newton's 3rd law. [1]

(iii) Momentum must be conserved [1]

because there is no external force acting on P and Q during the collision. [1]

However kinetic energy may or may not be conserved. [1]

It depends on whether the collision is elastic or not. [1]

(b) If the car is made of very strong material, it will be brought to rest in a very short time during a collision. [1]

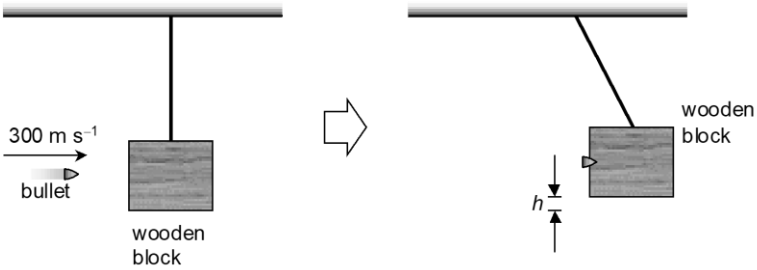
A large force then acts on the passengers which may cause serious injuries. [1]

14. A wooden block of mass 5 kg is hanging at rest. A bullet of mass 10 g moving at 300 m s^{-1} strikes the block horizontally. The bullet then embeds in the block and moves together with it as shown.

(a) Calculate the kinetic energy of the bullet before embedded in the block.

(b) Draw a diagram to show all the forces acting on the wooden block at the moment it is hit by the bullet.

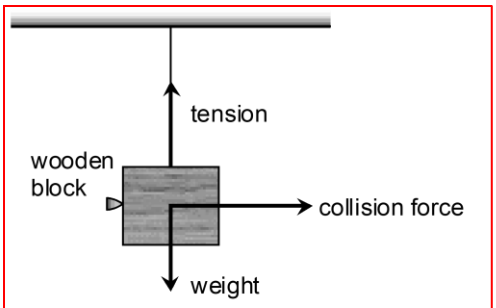
(c) How high would the block rise above its resting position?



(a) Applying $\mathrm{KE} = \frac{1}{2}mv^2$, the kinetic energy of the bullet is

$$\frac{1}{2} \times 0.01 \times 300^2 = 450\,\mathrm{J}.$$

(b)



(c) Consider the horizontal motions of the bullet and the block.

Let v_c be the common horizontal velocity of the block and the bullet.

By conservation of momentum, we have

$$0.01 \times 300 + 0 = (0.01 + 5)v_c$$
$$v_c = 0.5988\,\mathrm{m\,s^{-1}}$$ **(1M)**

The common horizontal velocity of the block and the bullet is 0.5988 m s^{-1}.

Applying $\mathrm{KE} = \frac{1}{2}mv^2$, the kinetic energy of the block and the bullet at this moment is

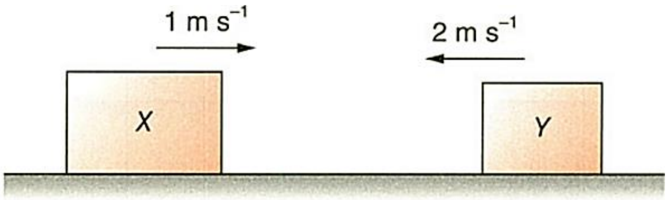
$$\frac{1}{2} \times (0.01 + 5) \times 0.5988^2 = 0.8982\,\mathrm{J}$$ **(1M)**

Assuming that all kinetic energy has been converted to potential energy. By conservation of energy, we have

$$(0.01 + 5) \times 9.81 \times h = 0.8982$$
$$h = 0.01828\,\mathrm{m}$$

The block would rise 0.0183 m or 1.83 cm.

15. Objects X and Y are moving on a horizontal smooth surface. X of mass 0.2 kg moves towards the right at $1\ \text{m s}^{-1}$. Y of mass 0.1 kg moves towards the left at $2\ \text{m s}^{-1}$. The objects collide at a certain instant. They move along a straight line before and after the collision.
Find the final velocities of the objects if the collision is
(a) completely inelastic, (2 marks)
(b) elastic (5 marks)



Take the direction to the right as positive.
(a) The objects have the same velocity after collision.
By conservation of momentum,
$m_X u_X + m_Y u_Y = (m_X + m_Y)v$ 1M
$0.2(1) + 0.1(-2) = (0.2 + 0.1)v$
$v = 0$ 1A
$\therefore$ The final velocities are both 0.

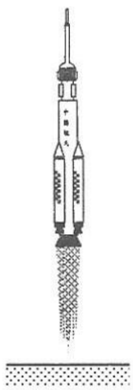
(b) By conservation of momentum,
$m_X u_X + m_Y u_Y = m_X v_X + m_Y v_Y$ 1M
$0.2(1) + 0.1(-2) = 0.2v_X + 0.1v_Y$
$v_Y = -2v_X$
By conservation of energy,
KE before collision = KE after collision 1M
$\frac{1}{2}(0.2)1^2 + \frac{1}{2}(0.1)2^2 = \frac{1}{2}(0.2)v_X^2 + \frac{1}{2}(0.1)v_Y^2$
$6 = 2v_X^2 + v_Y^2$
$= 2v_X^2 + (-2v_X)^2$ 1M
$v_X = -1\ \text{m s}^{-1}$ 1A
or $1\ \text{m s}^{-1}$ (rejected)
$\therefore$ The final velocity of X is $1\ \text{m s}^{-1}$ towards the left.
$v_Y = -2v_X$
$= -2(-1)$
$= 2\ \text{m s}^{-1}$ 1A
$\therefore$ The final velocity of Y is $2\ \text{m s}^{-1}$ towards the right.

16. A spacecraft with an astronaut on board is launched on a rocket booster. The rocket with the spacecraft has a total initial mass of 4.80×10^5 kg at take-off. The rocket engine propels hot exhaust gas at a constant speed of $2600\ \text{m s}^{-1}$ relative to the rocket in a backward direction. Assume that 2.30×10^3 kg of gas is expelled in the first second. (Neglect air resistance.)

(a) Calculate the average thrust (i.e. the upward force) acting on the rocket due to the exhaust gas during the first second. (2 marks)

(b) Assuming that the change of mass of the rocket during the first second is negligible, estimate the acceleration of the rocket. (2 marks)

(c) If the rocket keeps expelling exhaust gas at the same rate for the first 20 s, explain how the rocket's acceleration will change. (2 marks)



(a) $F = \frac{mv - mu}{t} = \frac{m}{t}(v - u) = (2.30\times10^3)(2600 - 0)$ [1]
$= 5.98 \times 10^6\ \text{N}$ [1]

(b) $F - mg = ma$ [1]
$(5.98 \times 10^6) - (4.80 \times 10^5)(9.81) = (4.80 \times 10^5)\,a$
$\therefore\ a = 2.65\ \text{m s}^{-2}$ [1]

(c) Since the average thrust remains unchanged, as the mass of the rocket gradually decreases, [1]
the acceleration of the rocket would gradually increase. [1]

17. A golf ball, of mass 40 g and initially at rest, is struck with a club in teeing off (see Figure 14). The ball leaves the club with a speed $44\ \text{ms}^{-1}$. Assume that air resistance is negligible.

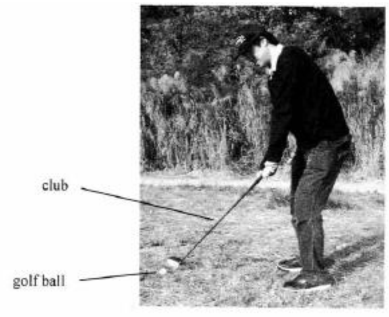


Figure 14

(a) (i) Calculate the change in momentum of the golf ball before and after teeing off. (2 marks)

(ii) The time of impact between the club and the ball during teeing off is 1 ms. Determine average force acting on the ball during the impact. (2 marks)

(b) Robert finds that the club is harder than the golf ball. He claims that the force exerted on the club is smaller than that exerted on the golf ball during teeing off. Explain whether his claim is correct or not. (2 marks)

(c) When the golf ball is 2.5 m away from the hole, it is given a sharp horizontal push from rest and just reaches the hole (see Figure 15). Estimate the initial speed of the golf ball if the average resistive force exerted on the ball is 0.03 N. (3 marks)

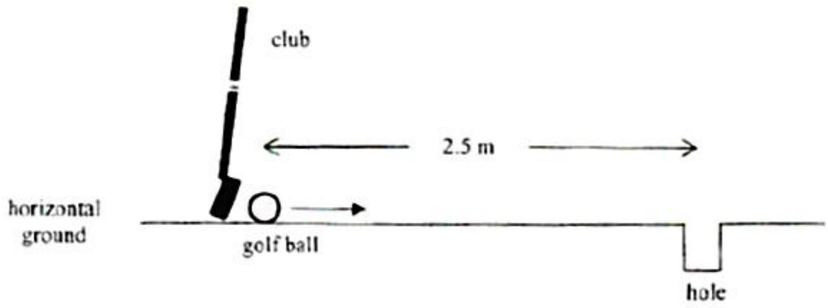


Figure 15

(a) (i) change in momentum $= mv - mu$	1 M
$= (0.04)(44) - 0$	
$= 1.76$ N s	1 A

(ii) force $= \dfrac{mv - mu}{t}$

$= \dfrac{1.76}{0.001}$

$= 1760$ N

(b) No. Both the club and the ball received the force of the same magnitude as they are action and reaction pair of forces.	1 A 1 A

(c) $fs = \frac{1}{2}mu^2$	1 M+1 M
$(0.03)(2.5) = \frac{1}{2}(0.04)u^2$	
$u \approx 1.94\ \text{m s}^{-1}$	1 A

18. A trolley of mass 2 kg travels with a constant velocity of 0.5 m s^{-1} to the right on a smooth horizontal ground. It collides with a spring attached to a wall fixed to the ground as shown. The trolley rebounds with the same speed.

(a) (i) Draw an action-and-reaction pair on the system during the collision. (1 mark)

(ii) By Newton's third law, the action force and the reaction force are equal in magnitude and opposite in direction. State another property about these forces. (1 mark)

(iii) State an application of Newton's third law. (1 mark)

(b) Sketch a graph to show how the momentum of the trolley varies with time for the above collision. Take the direction to the right as positive. (2 marks)

(c) If the impact time during the collision is 0.5 s, calculate the average impact force acting on the trolley. (2 marks)

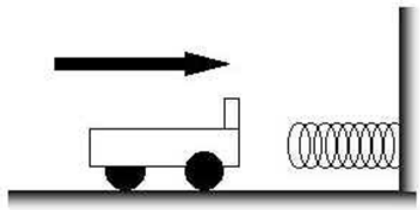
(d) Find the kinetic energy of the trolley just before the collision. (2 marks)

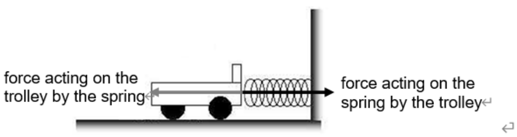
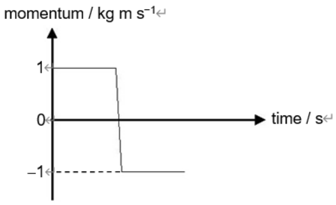
(e) Two students make the following remarks about the experiment.

John says, 'The momentum of the trolley is conserved in the collision.'

Peter says, 'When the trolley comes to a stop momentarily during the collision, the energy of the above system disappears.'

Explain whether each of the above remarks is correct. (4 marks)



(a) (i)	
	
(Correct action-and-reaction pair)	1A
(ii) They act on the different bodies.	1A
(iii) Rockets fly upwards by ejecting gas downwards.	1A
(b)	
	
(Correct labelled axes with units)	1A
(Correct graph)	1A
(c) Average impact force $= \frac{mv - mu}{t}$	1M
$= \frac{2 \times (-0.5) - 2 \times 0.5}{0.5}$	
$= -4$ N	
The average impact force is 4 N towards the left.	1A
(d) Kinetic energy of the trolley $= \frac{1}{2} mv^2$	1M
$= \frac{1}{2} \times 2 \times 0.5^2$	
$= 0.25$ J	1A
(e) John is not correct.	1A
The momentum of the trolley changes before and after the collision as there is a net force acting on it.	1A
Peter is not correct.	1A
The kinetic energy of the trolley is converted into the elastic potential energy stored in the spring.	1A